Ultrasound and optoacoustic imaging
Part III: Optoacoustic imaging

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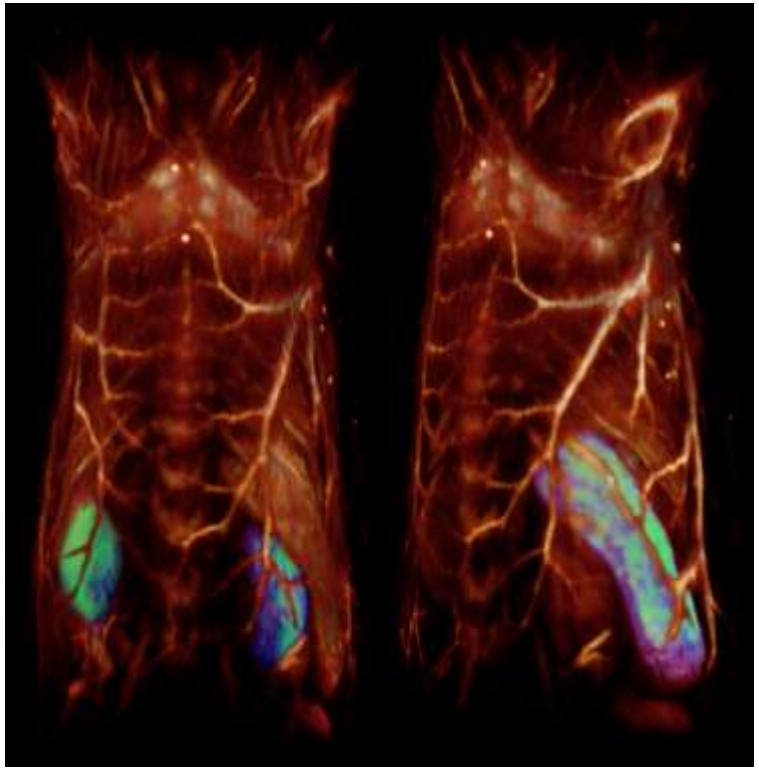
23.04.2026

Today's topic



Learning objectives

Part I: Ultrasound in biology and medicine

Part II: Advanced ultrasound imaging

Part III: Optoacoustic imaging

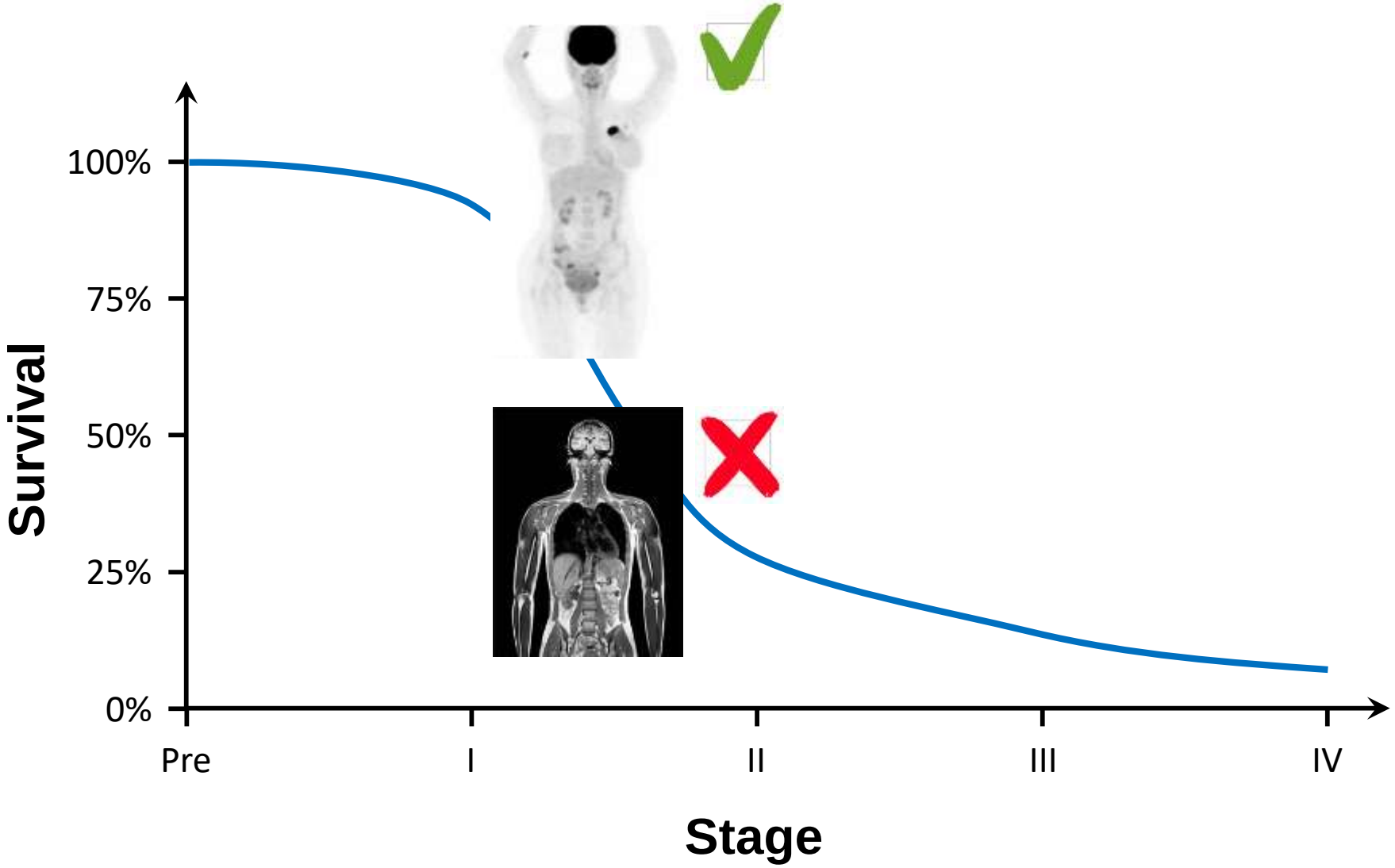
What do we see in an image?

Example: what do you see?

Structural imaging is important…

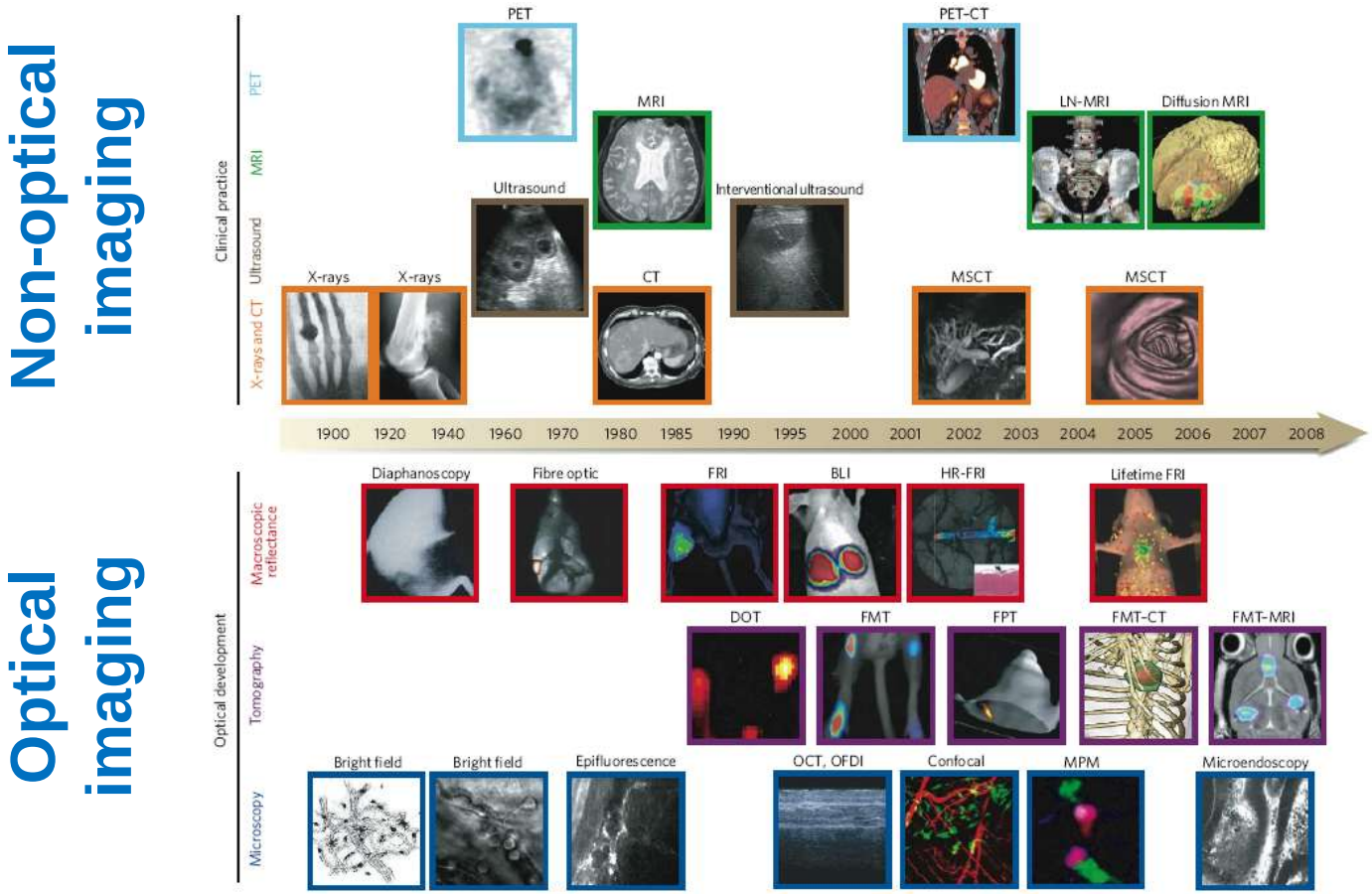
…but contrast specificity is also helpful

The importance of early diagnosis



Optical imaging

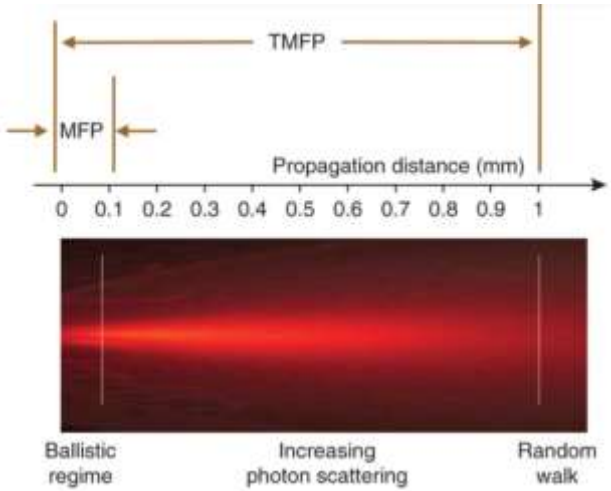
Optical imaging encompasses multiple modalities exploiting photon-molecule interactions at ultraviolet to infrared wavelengths



Weissleder R. and Pittet M. J. Imaging in the era of molecular oncology. Nature 452:580-9 (2008)

The deep-tissue challenge

Photons undergo **scattering** events as they propagate through biological tissues

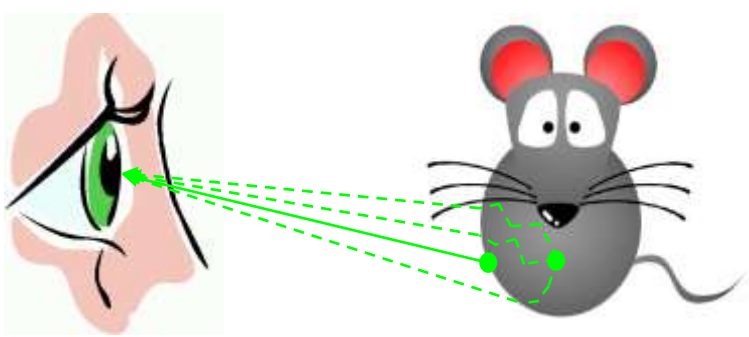


Ntziachristos V. Going deeper than microscopy: the optical imaging frontier of biology. Nat. Methods 7:603-614 (2010)

MFP (mean free path) – average distances a photon travels between scattering events

TMFP (transport mean free path) – average distance for which photon propagating distances are uncorrelated

The achievable resolution rapidly decays with depth, particularly at depths beyond 1mm where light becomes diffusive

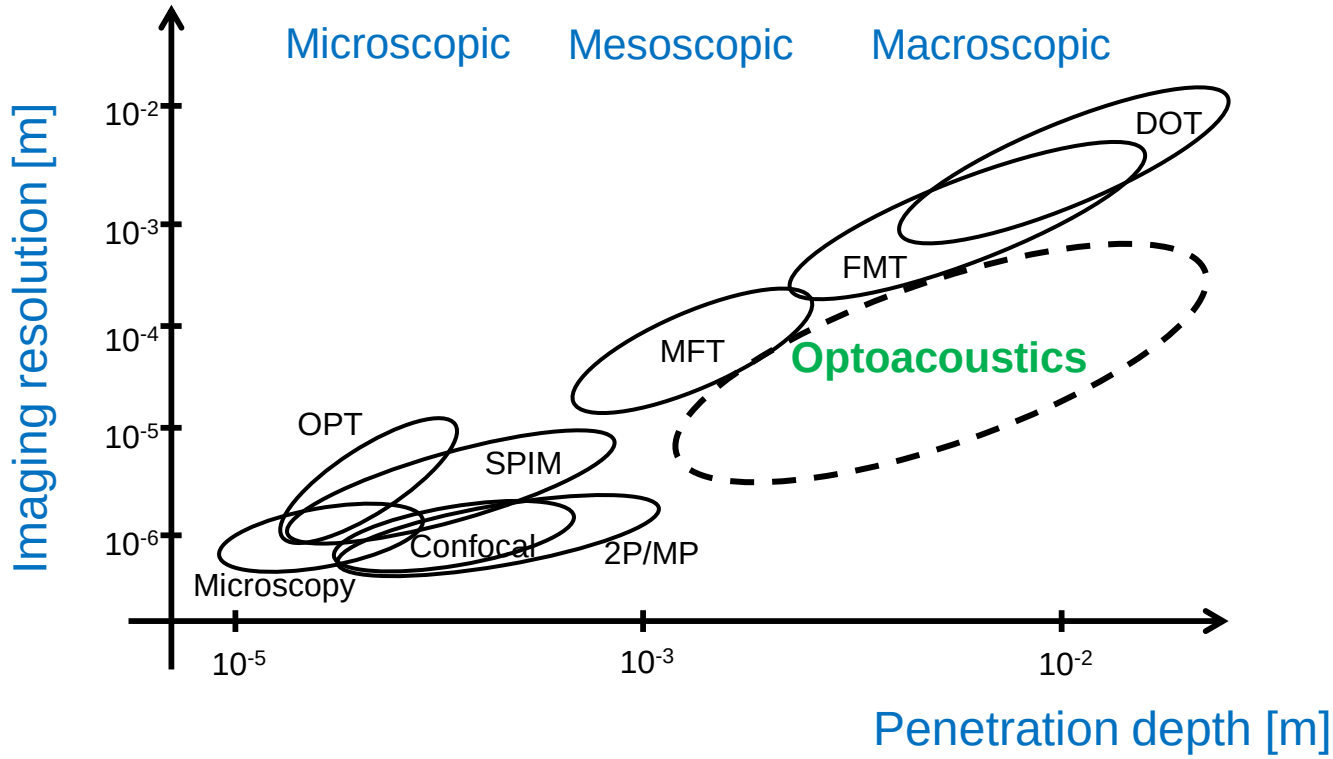


scattering of light (fog)

Resolution and depth

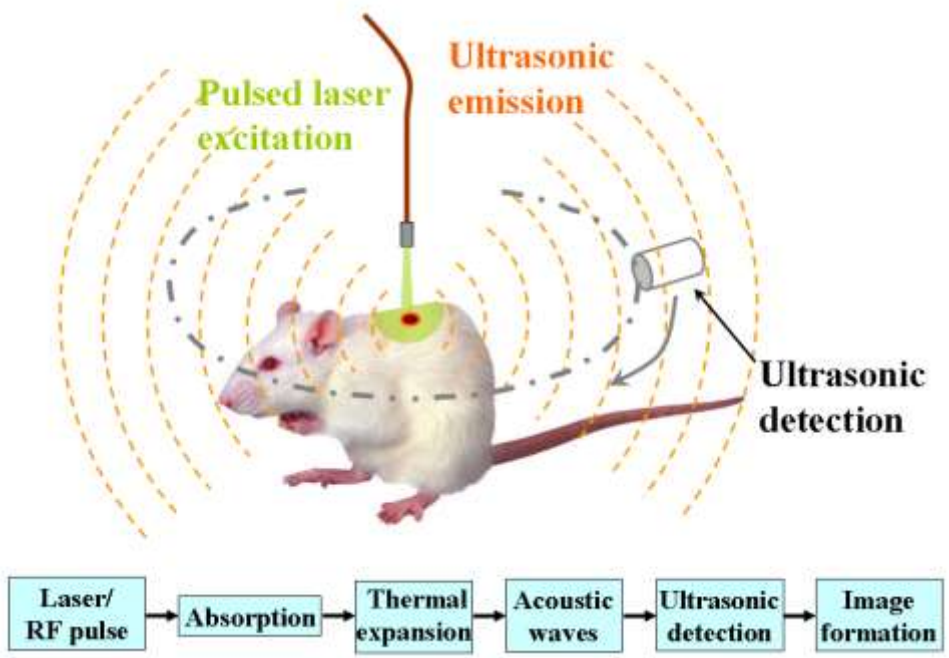
Diffuse optical imaging modalities typically provide a spatial resolution in the order of 1/10 of the imaging depth

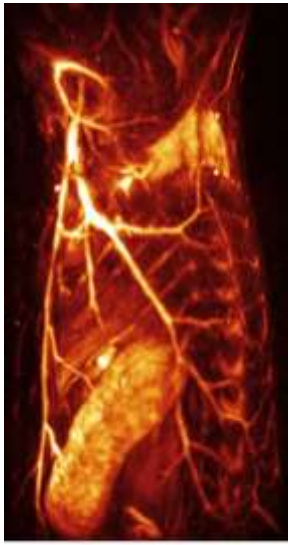
Optoacoustic imaging typically provides a spatial resolution in the order of 1/200 of the imaging depth



Optoacoustic (photoacoustic) imaging

Optoacoustic imaging (also known as photoacoustic imaging) synergistically combines the advantages of light and ultrasound to provide optical contrast at millimetre to centimetre scale depths



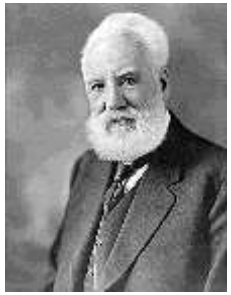


Light In

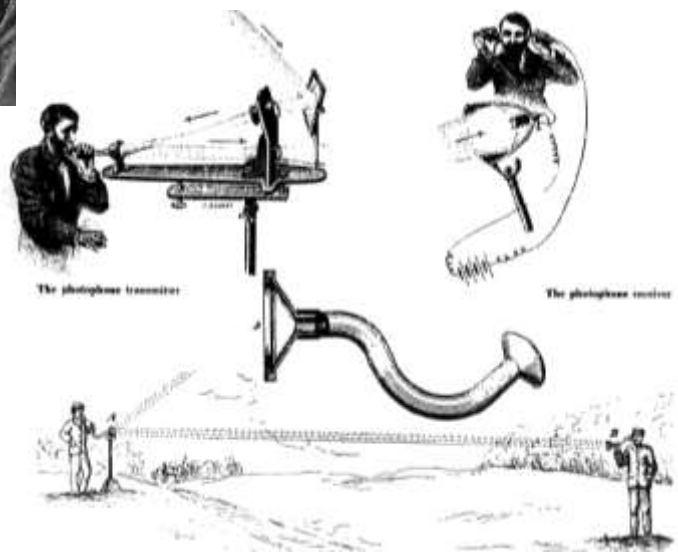
Ultrasound Out

From the principle to in vivo imaging

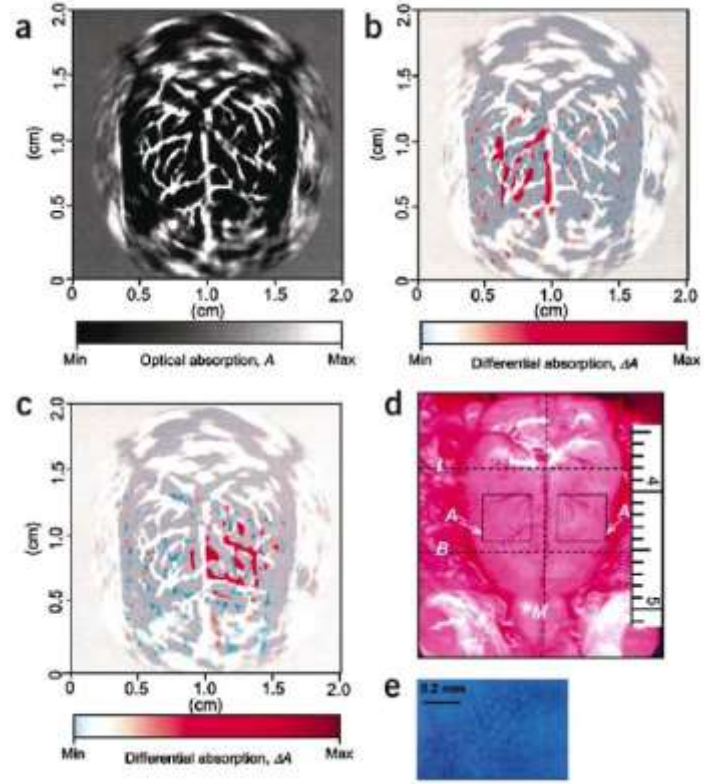
The “photophonic” principle was first discovered at the end of the 19th century while the first functional in vivo images were reported at the beginning of the 21st century



Alexander Graham Bell



Tainter and Bell. Selenium and the photophone. Nature; 22: 500–503 (**1880**)



Wang et al. Noninvasive laser-induced photoacoustic tomography for structural and functional in vivo imaging of the brain. Nature biotechnology, 21(7), 803-806 (**2003**)

A bit of history

Optoacoustics has a long history but only recently it has been developed as a biomedical imaging modality

- 1880 – Observation of the optoacoustic (photophonic) effect (Alexander Graham Bell)

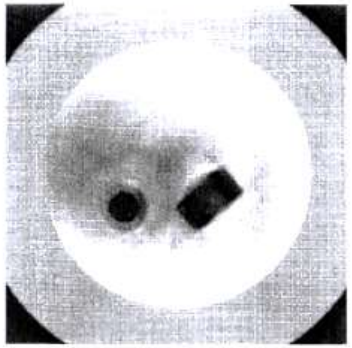
- 1960 – Invention of the laser (Theodore Maiman)

- 1981 – Proposed thermoacoustic imaging of soft tissues (Theodore Bowen)

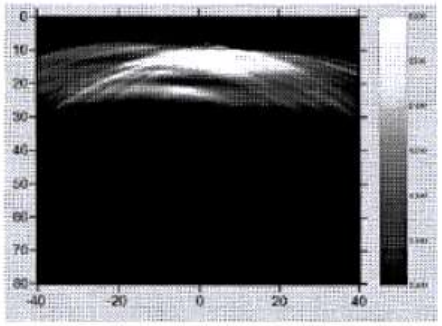
- 1995 – First optoacoustic image of a tissue phantom (Robert Kruger)

- 1999 – First *in vivo* optoacoustic image (Alexander Oraevsky)

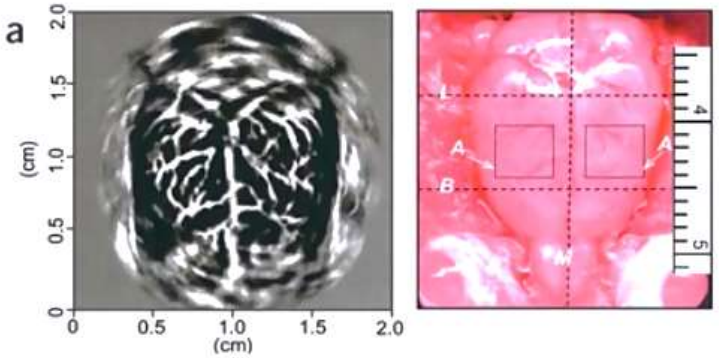
- 2003 – First *in vivo* functional optoacoustic image (Lihong Wang)



Kruger et al. (1995) Med. Phys. 22, 1605-1609



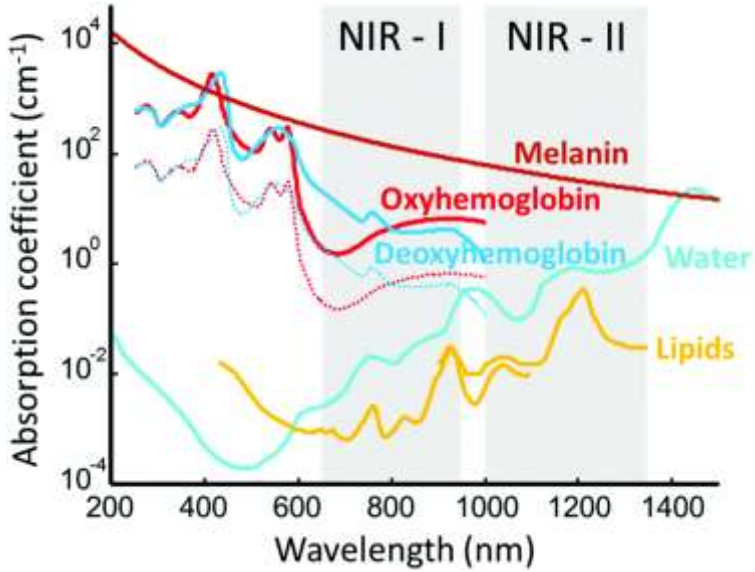
Oraevsky et al. (1999) Proc SPIE 3597, 352-363

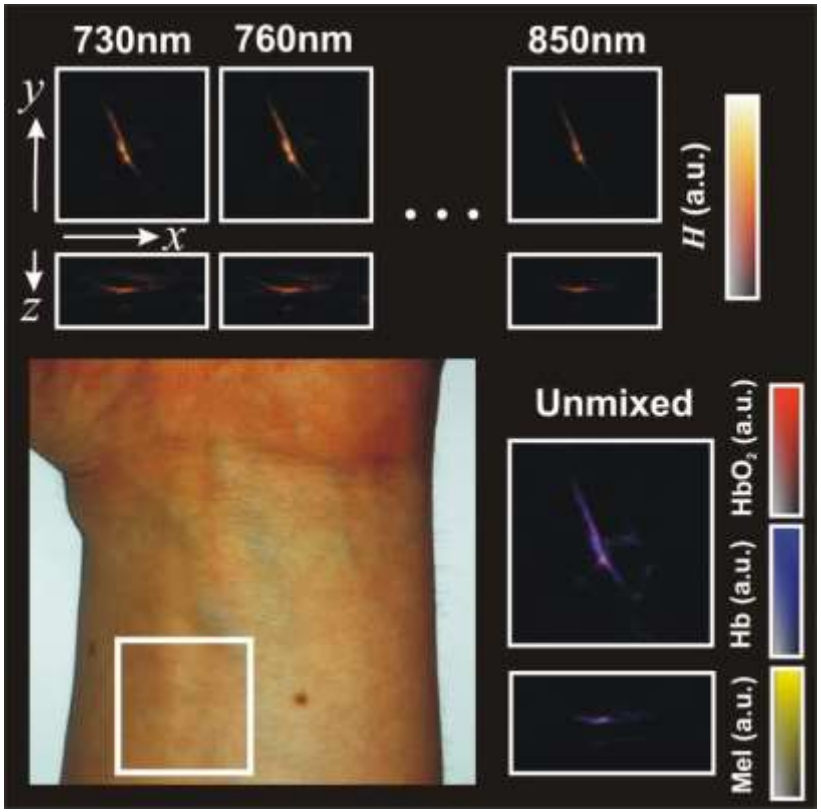


Wang et al. (2003) Nat. Biotech., **21**, 803 – 806.

Five dimensional imaging

Multi-spectral optoacoustic tomography (MSOT) is generally performed in the near-infrared window to maximize the achievable depth

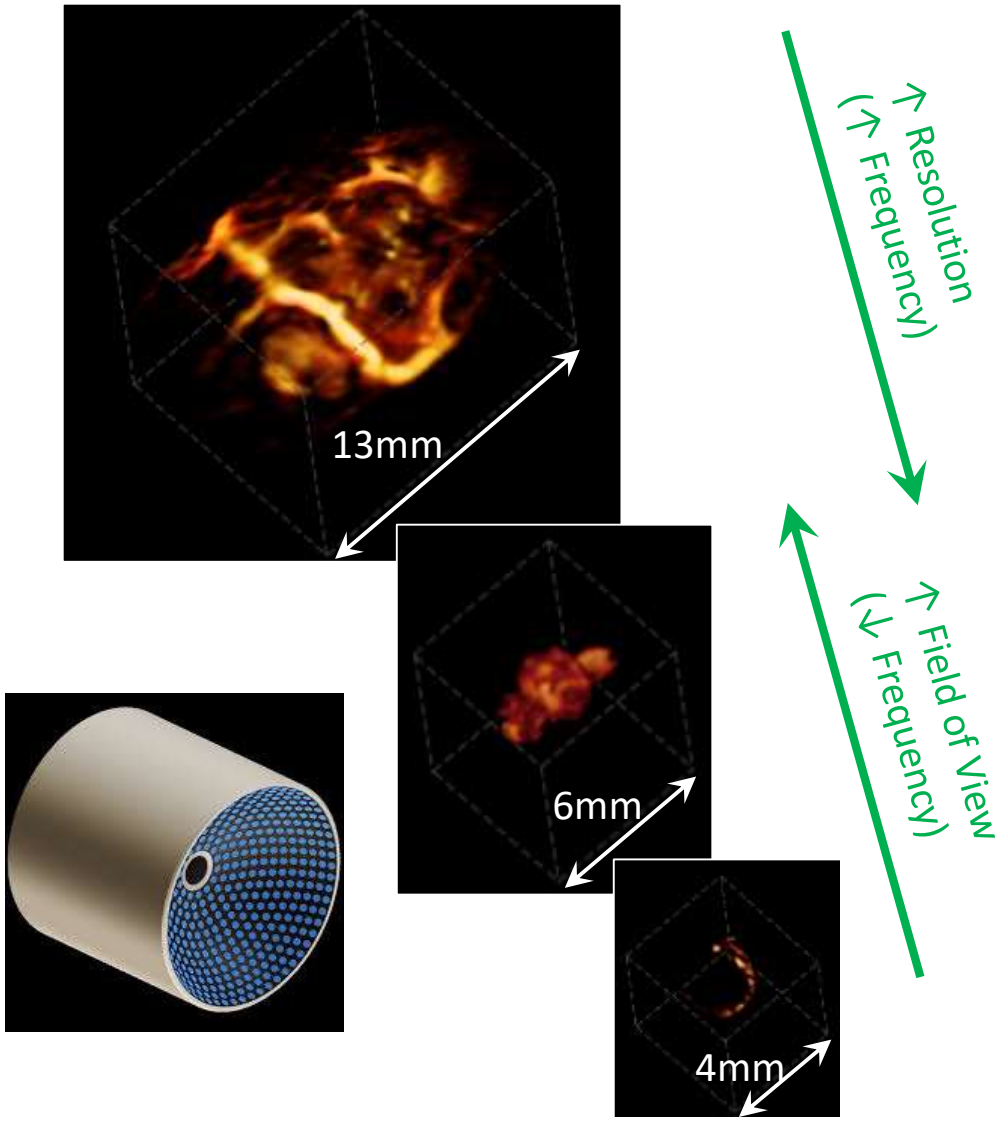




Three spatial dimensions, time, and optical wavelength represent **five independent dimensions**

Multi-scale imaging

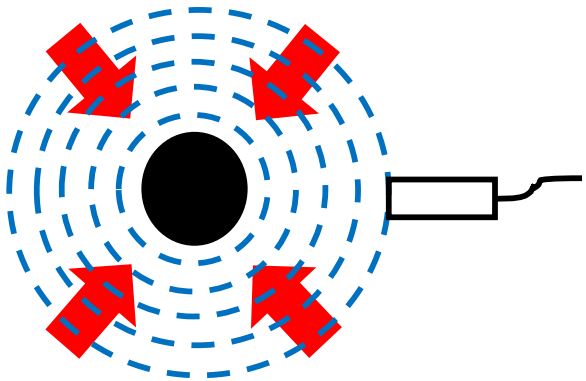
The field of view and spatial resolution are scalable with frequency

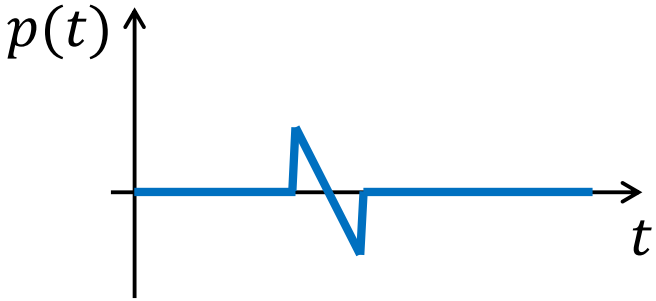


The field of view, spatial and temporal resolution can be adapted to the application of interest

Optoacoustic signal generation

Optoacoustic signals are generally generated by short pulsed laser pulses (<100ns, typically ~10ns)





Optoacoustic contrast corresponds to the induced **initial pressure rise**, proportional to the **absorbed energy** per unit volume

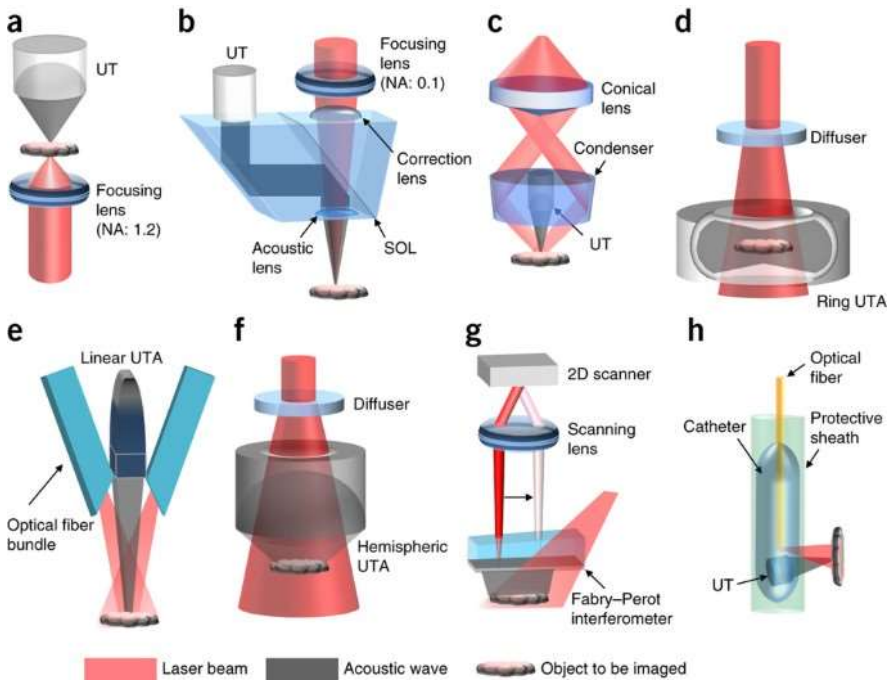
$$p_0 = \frac{\beta}{\kappa_T C_V} \mu_a \phi$$

Grueneisen parameter

Absorbed energy

Optoacoustic embodiments

Multiple types of optoacoustic hardware systems (embodiments) are available to cover different spatial and temporal scales



Wang and Yao. A practical guide to photoacoustic tomography in the life sciences. Nature methods, 13(8), 627-638 (2016)

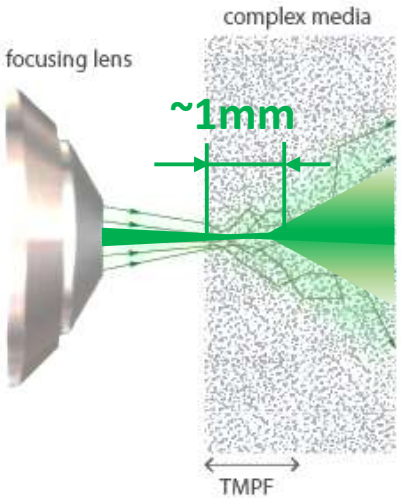
Optoacoustic systems are generally classified into two categories

- **Optoacoustic microscopy**
- **Optoacoustic tomography**

Optoacoustic microscopy

Optical-resolution optoacoustic microscopy

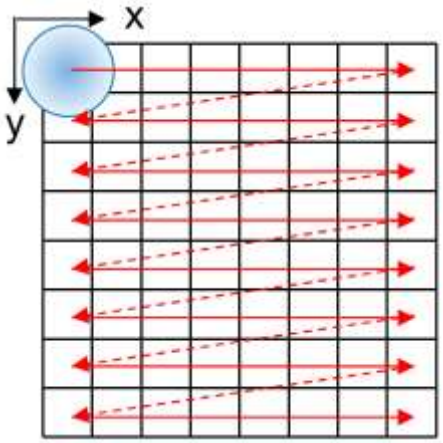
In **optical-resolution optoacoustic microscopy** (OR-OAM), directionality is achieved with a laser beam

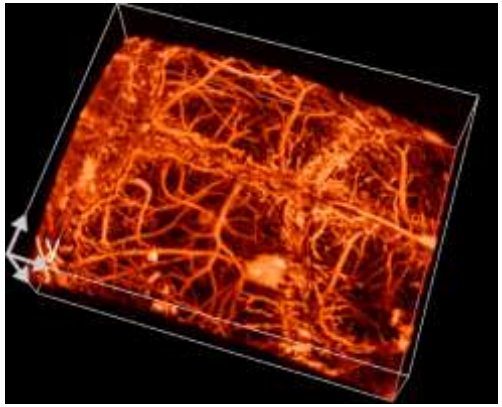


Optical focusing is possible for <1mm depth

Lateral resolution is established by the diameter of the light spot

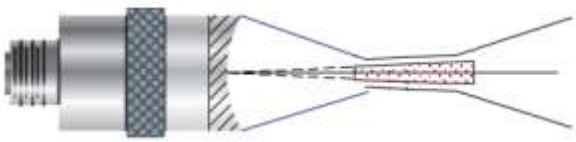
Absorbers at different depths can be distinguished with a high-bandwidth single-element transducer





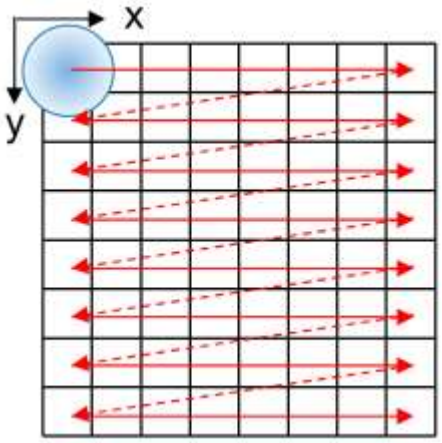
Acoustic-resolution optoacoustic microscopy

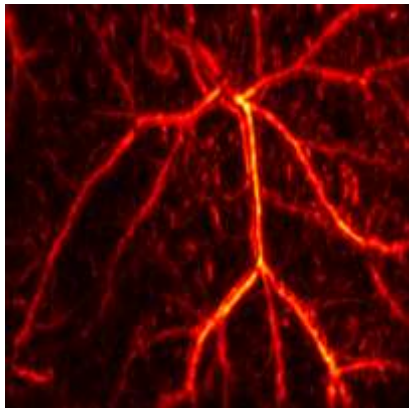
In **acoustic-resolution optoacoustic microscopy** (AR-OAM), lateral selectivity is achieved with a focused transducer



The lateral resolution is limited by acoustic diffraction

The axial resolution is limited by the bandwidth of the transducer as in OR-OAM





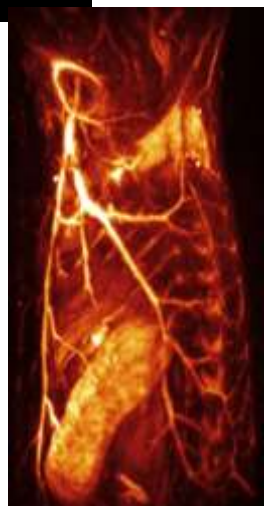
Optoacoustic tomography

The tomographic nature of optoacoustics

The optoacoustic Poisson-type integral (forward model) can be inverted to derive a **back-projection formula** in a similar manner as the Radon transform in x-ray CT

X-ray CT

$$p_\theta(\boldsymbol{r}) = -\int_{L(\boldsymbol{r},\theta)} \mu(\boldsymbol{r}')d\boldsymbol{r}' \longrightarrow \mu(\boldsymbol{r}') = \int_0^\pi Q_{\theta-\frac{\pi}{2}}(r)d\theta$$



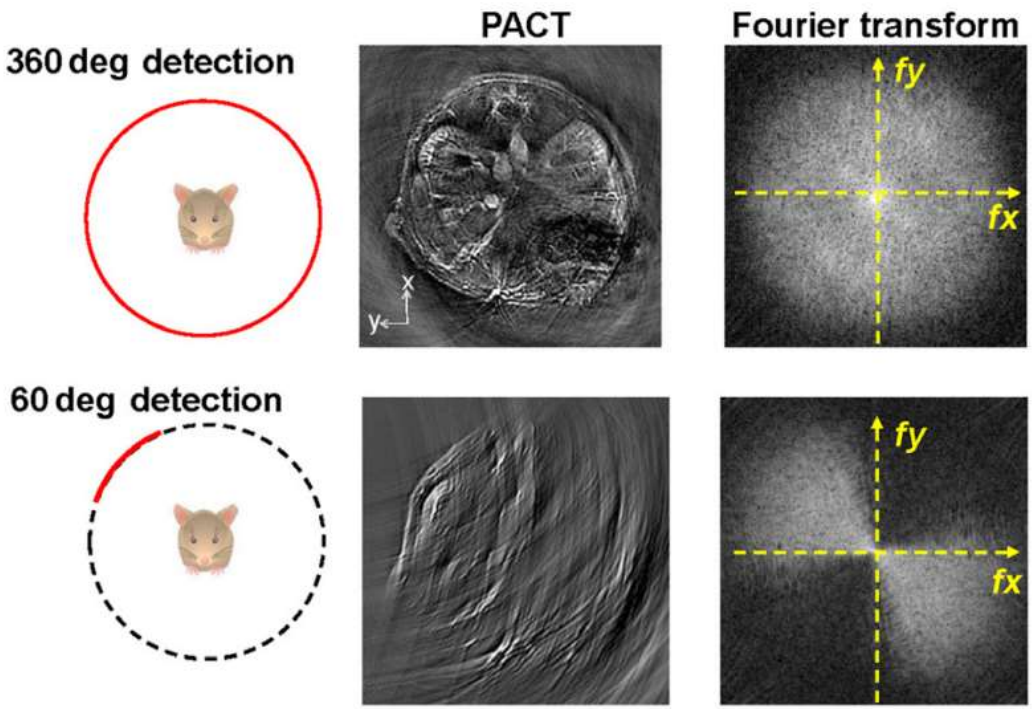
Optoacoustics

$$p(\boldsymbol{r},t) = \frac{\partial}{\partial t}\frac{1}{t}\int_{S(\boldsymbol{r},t)} p_0(\boldsymbol{r}')d\boldsymbol{r}' \longrightarrow p_0(\boldsymbol{r}') = \int_\Omega \frac{d\Omega}{\Omega}[s(\boldsymbol{r},t)]_{t=|r-r'|/c}$$

$$s(\boldsymbol{r},t) = 2p(\boldsymbol{r},t) - 2t\frac{\partial p(\boldsymbol{r},t)}{\partial t}$$

Limited-view effects

Much like for other tomographic imaging modalities, accurate optoacoustic images can only be obtained if signals from different positions around the sample are collected



Limited-view artefacts are produced if the angular coverage is insufficient (**<180°**)

Spatio-temporal resolution

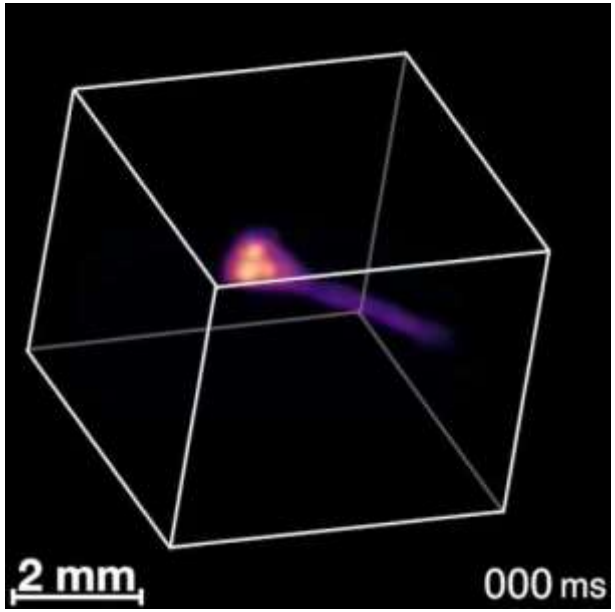
Optoacoustic tomography covers a large spatio-temporal resolution range and larger depths than optoacoustic microscopy

Spatial resolution

For full-tomographic coverage, the spatial resolution is isotropic

Temporal resolution

The temporal resolution is established by the pulse repetition rate of the laser



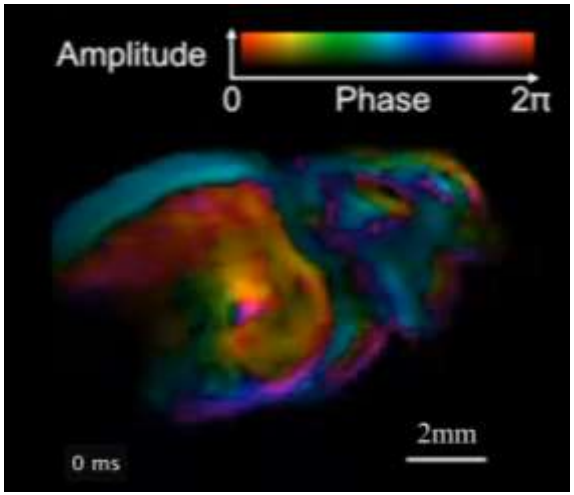
Model-based reconstruction

Model-based optoacoustic reconstruction is possible by discretizing the forward optoacoustic model

$$p(\boldsymbol{r},t) = \frac{\partial}{\partial t}\frac{1}{t}\int_{S(\boldsymbol{r},t)} p_0(\boldsymbol{r}')d\boldsymbol{r}' \longrightarrow \mathbf{p} = \mathbf{A}\mathbf{p}_0$$

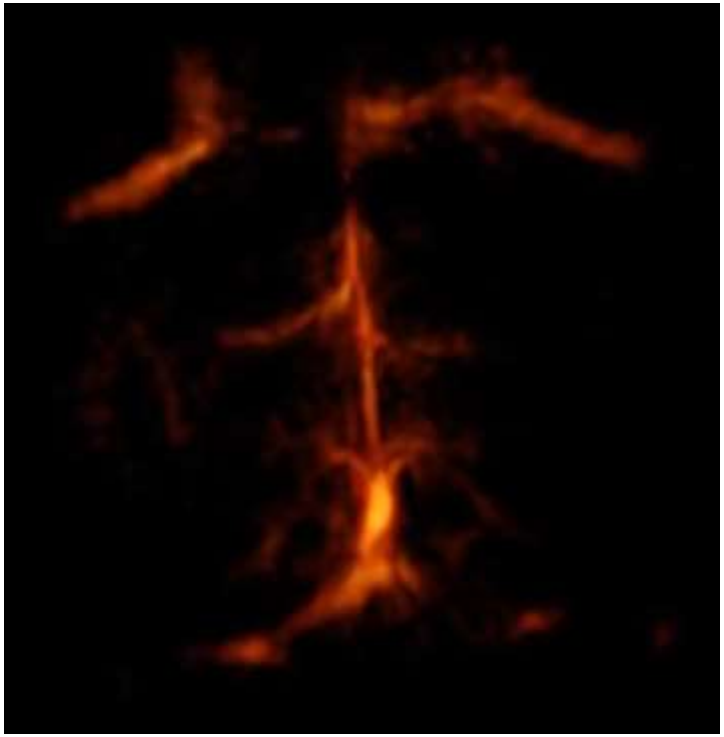
Different types of regularization are possible, including ad-hoc approaches to simultaneously reconstruct a sequence of images

$$\mathbf{p}_{0,\mathrm{sol}} = \underset{\mathbf{p}_0}{\mathrm{argmin}}\{\|\mathbf{A}\mathbf{p}_0 - \mathbf{p}\| + R(\mathbf{p}_0)\}$$



Optoacoustic super-resolution imaging

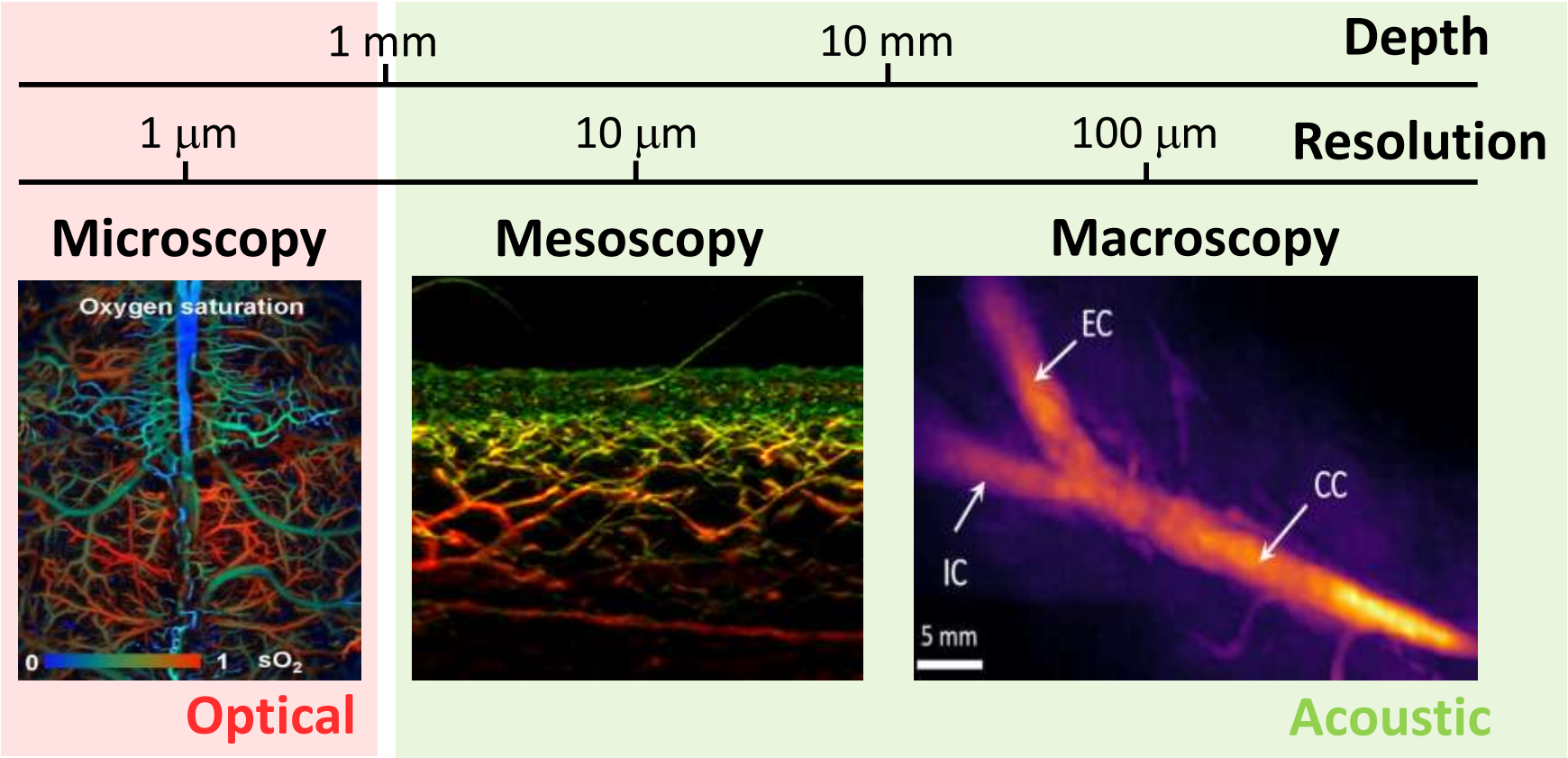
Localization optoacoustic tomography (LOT) for functional super-resolution imaging



Clinical translation

Multi-scale imaging

Optoacoustics can cover microscopic to macroscopic scales with different embodiments



The typical spatial resolution is in the order of 1/200 of the depth

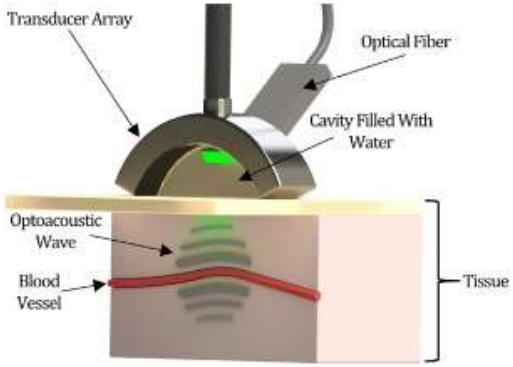
Hand-held optoacoustic imaging

Optoacoustic imaging can be performed with newly-developed hald-held probes operated in the same manner as standard pulse-echo ultrasound scanners

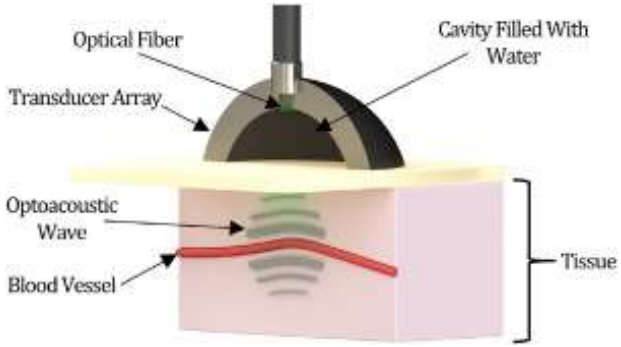
Concave hand-held scanners

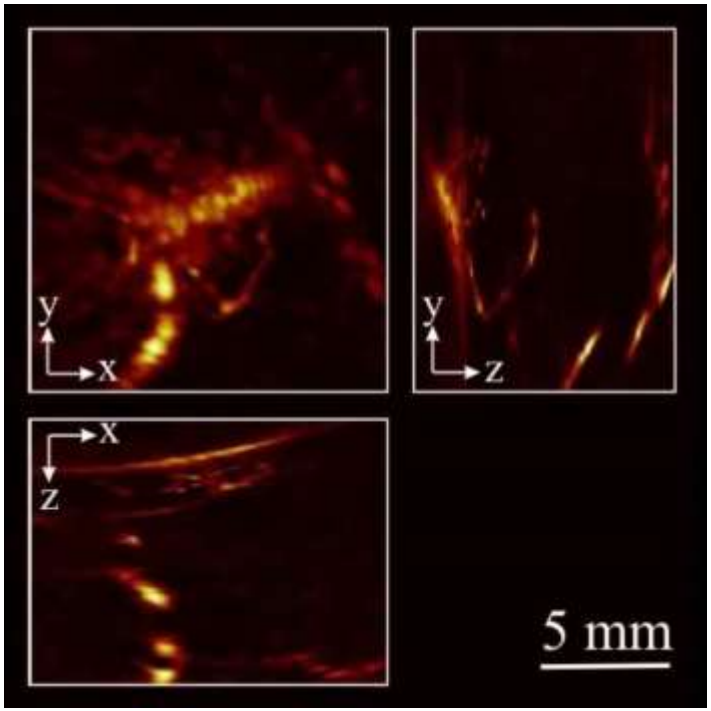
Custom-made concave arrays are preferred over linear (or planar) arrays to maximize the tomographic angular coverage

Cylindrically-focused array



Spherical array





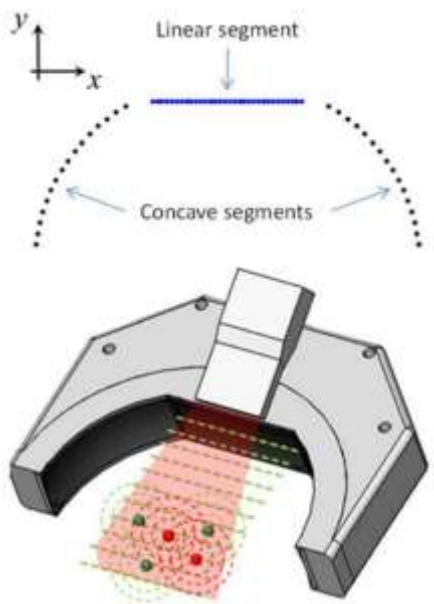
Optoacoustics + ultrasound

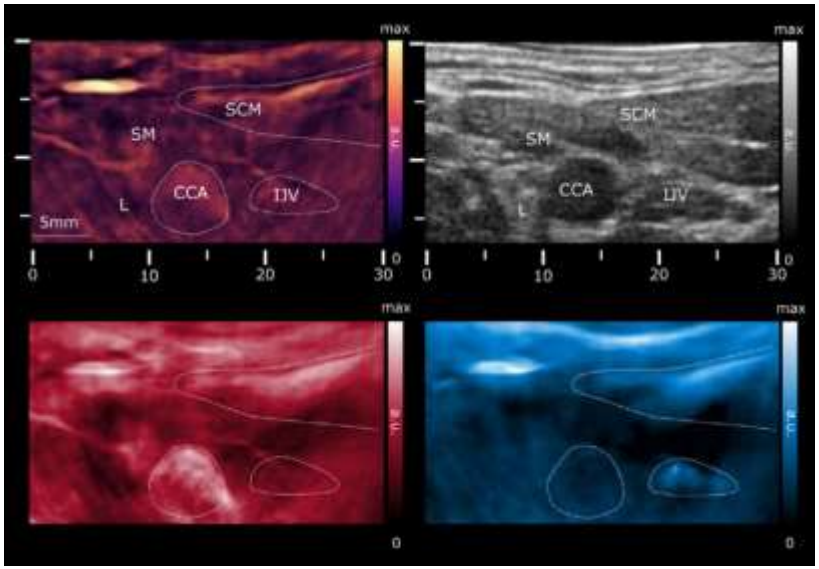
Optoacoustic and **pulse-echo ultrasound imaging** are “natural partners” as both are based on the detection of ultrasound signals with arrays of transducers

Pulse-echo ultrasound – linear or convex arrays with small inter-element pitch

Optoacoustics – concave arrays with large angular coverage

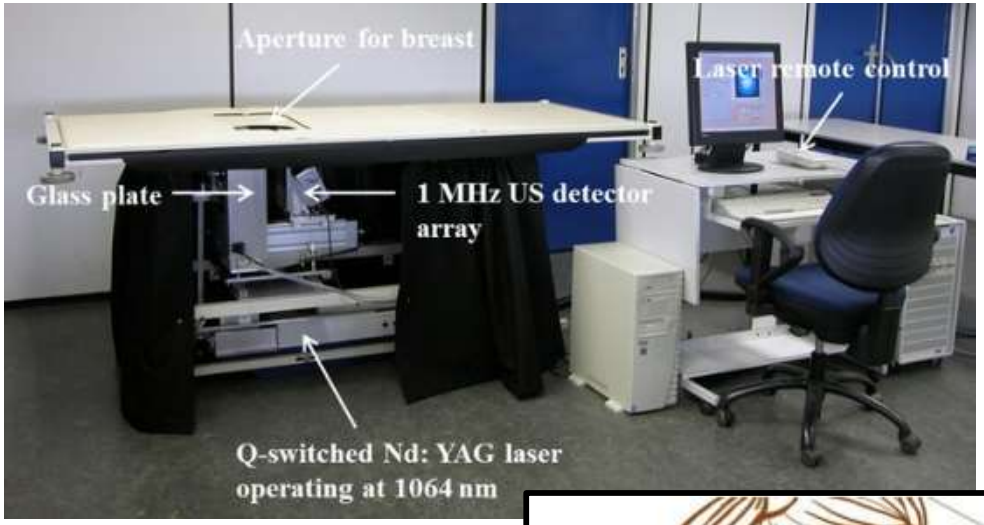
Example of hybrid solution: Multi-segment array

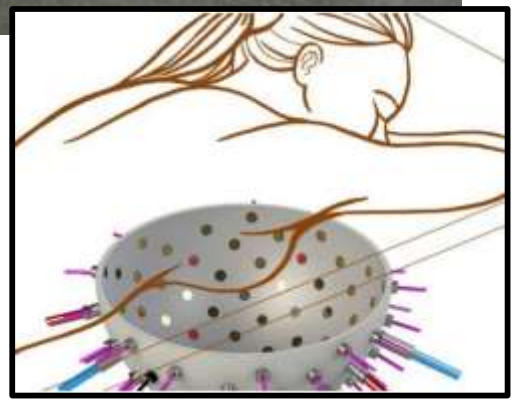


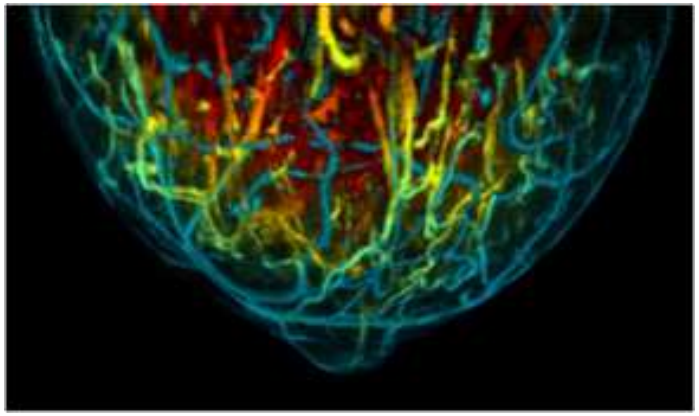


Tomographic systems

Custom-made optoacoustic tomographic imaging systems have been developed to cover the entire female breast, which has reduced optical absorption compared to other tissues

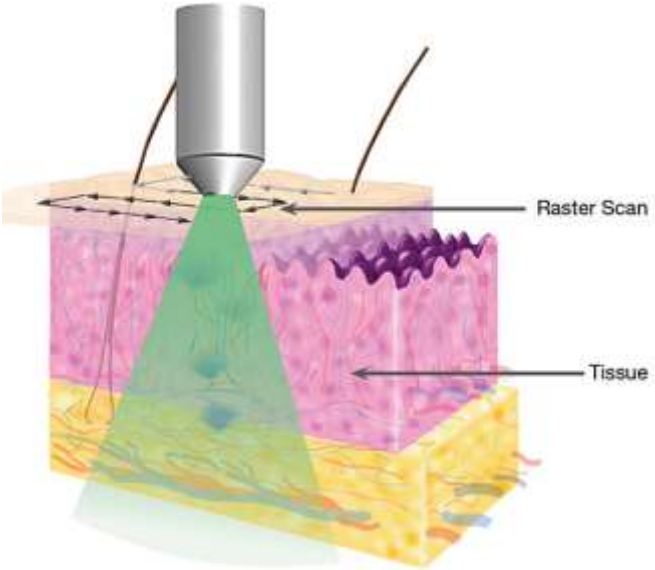






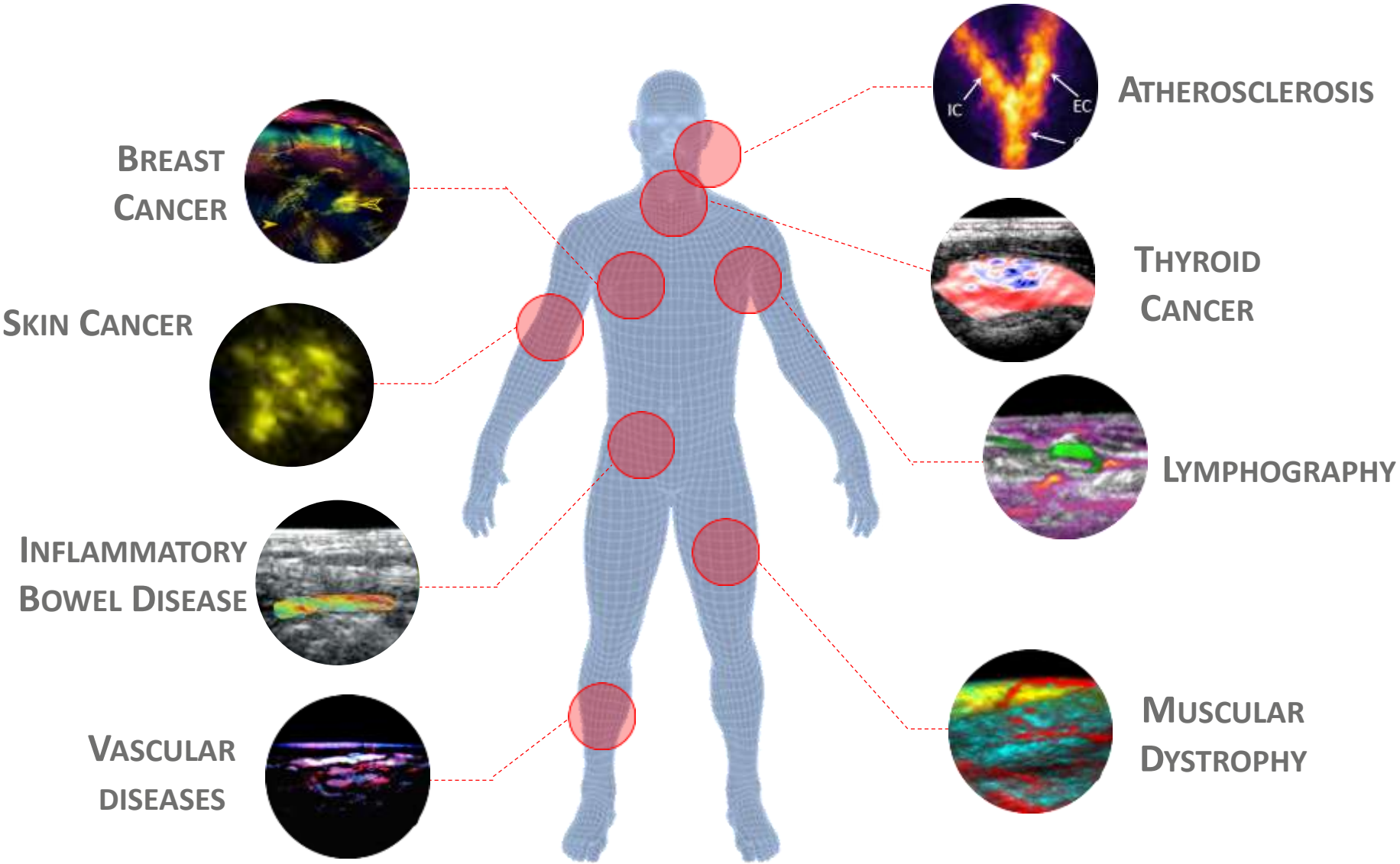
Raster-scan optoacoustic imaging

Acoustic-resolution optoacoustic mesoscopy systems have been developed for clinical use, particularly to image the skin



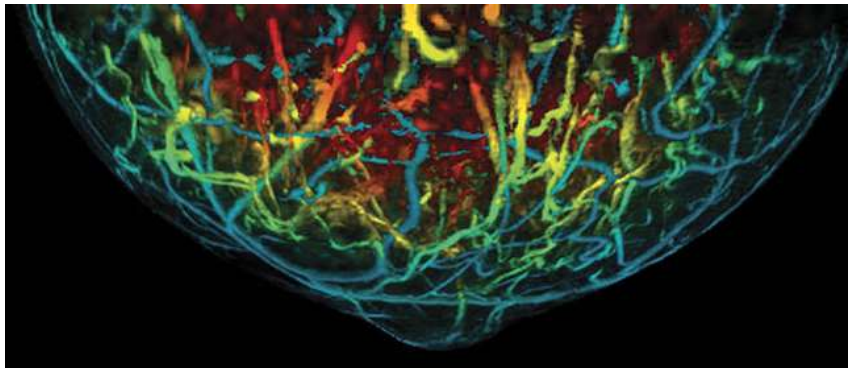
Clinical applications

Optoacoustic clinical trials

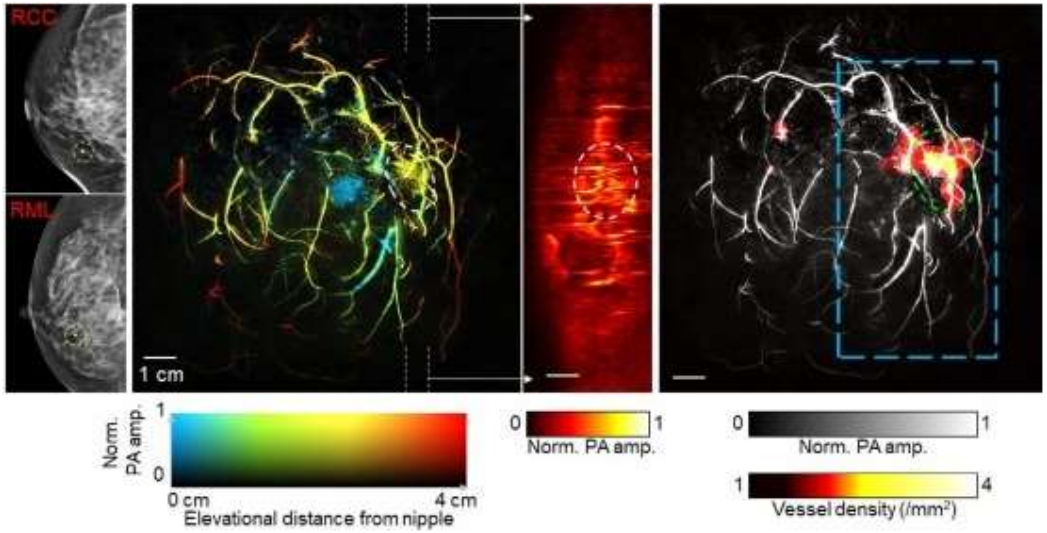


Breast cancer

Light absorption and scattering in the female breast is lower than in other tissues, which facilitates reaching deep regions

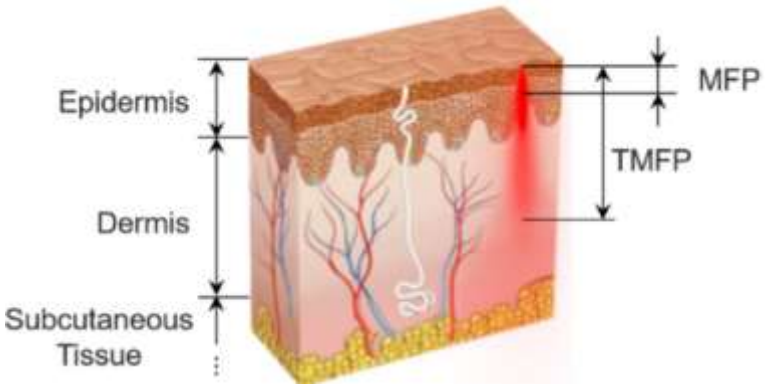


Optoacoustics has been shown to be capable of visualizing vascular changes in breast tumors

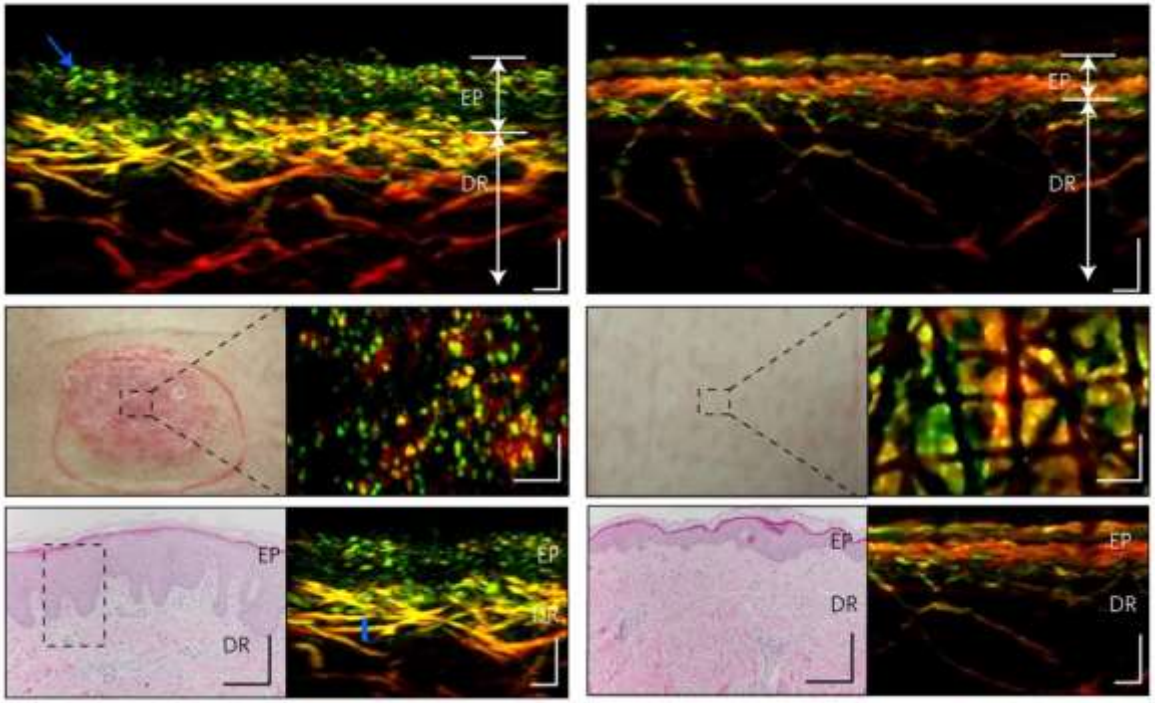


Skin conditions

The skin (maximum thickness 2-3mm) is easily accessible with optoacoustic microscopy and tomography systems

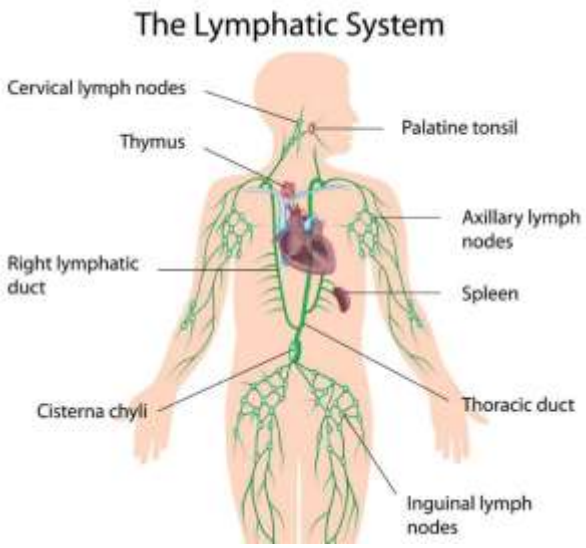


Example Assessment of psoriasis

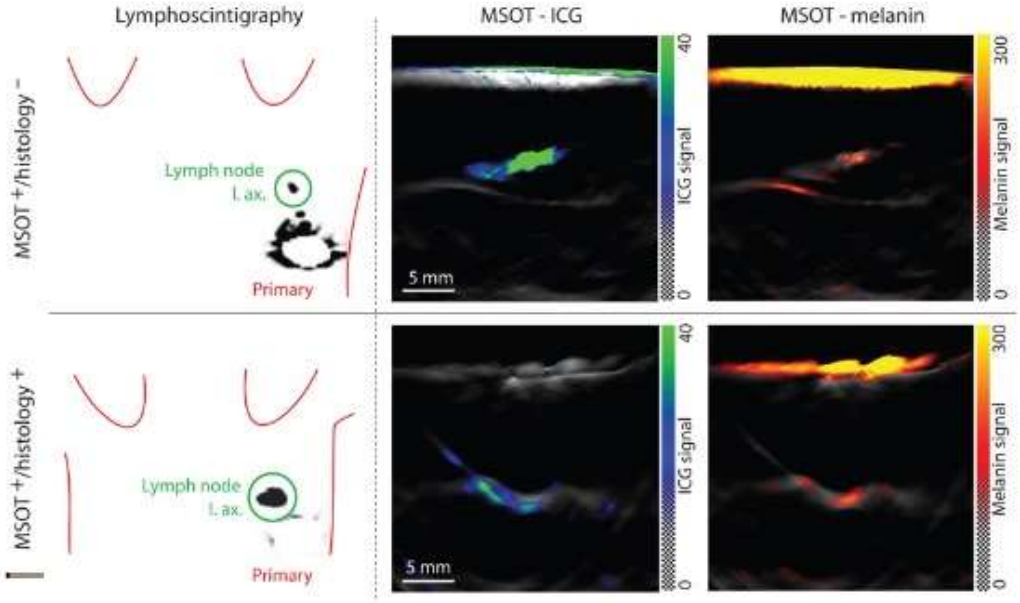


Lymph node metastasis

Lymph nodes are the most common sites of metastasis in cancer patients



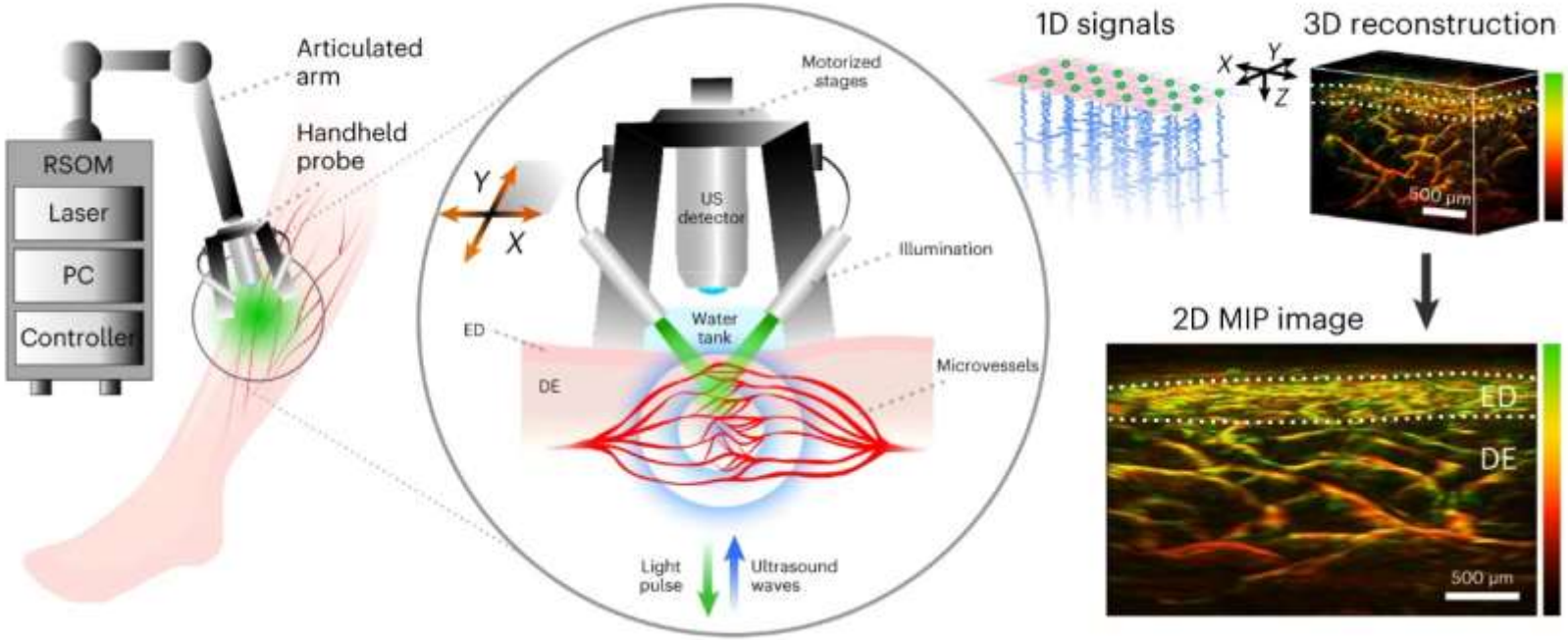
Optoacoustics provides sufficient penetration depth to reach some lymph nodes



Diabetes

Diabetes mellitus is a group of common endocrine diseases characterized by sustained high blood sugar levels

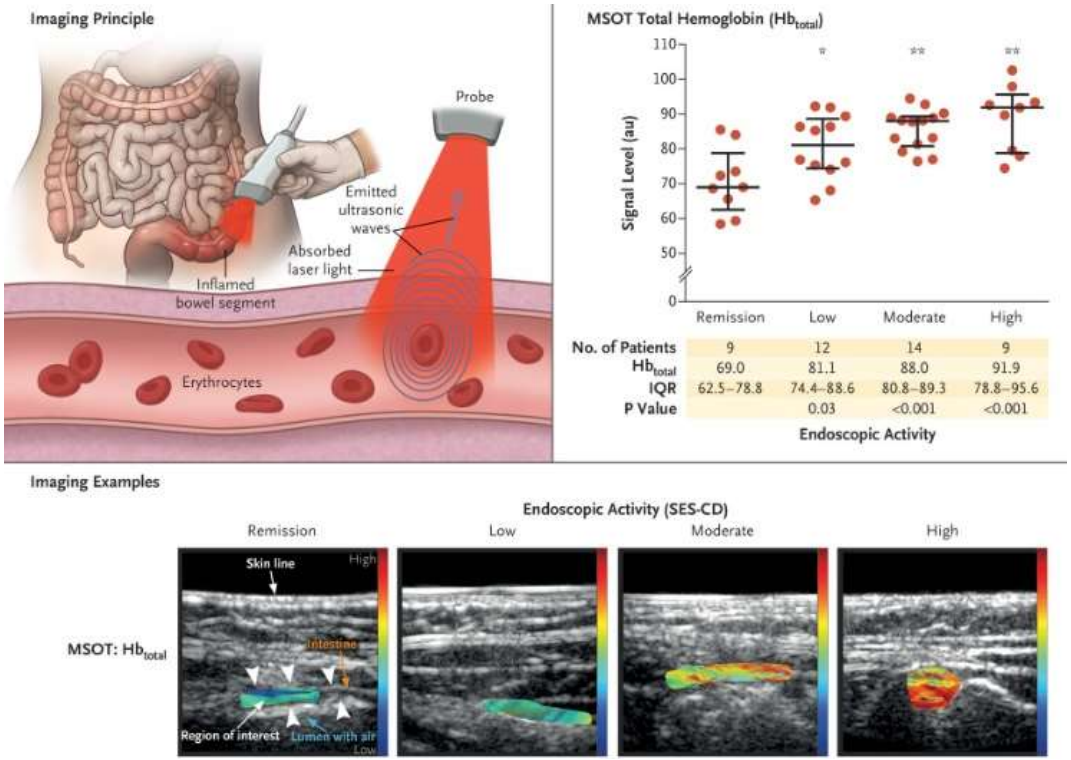
Skin microangiopathy is associated with diabetes, which can be characterized with optoacoustics



Crohn disease

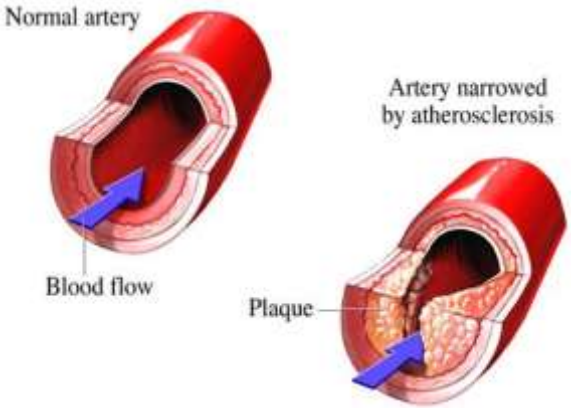
Crohn's disease is a type of chronic inflammatory bowel disease associated with increased rates of hospitalization and death

Optoacoustics could visualize changes in hemoglobin in Crohn's disease patients

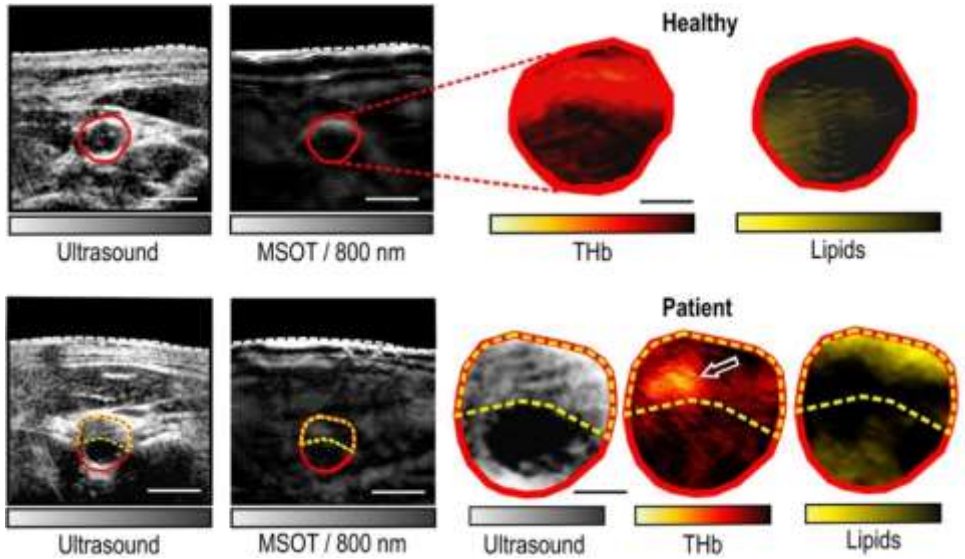


Atherosclerotic plaques

Atherosclerosis refers to an inflammatory build-up of atheromatous plaque inside the common carotid artery, or the internal carotid arteries that causes them to narrow



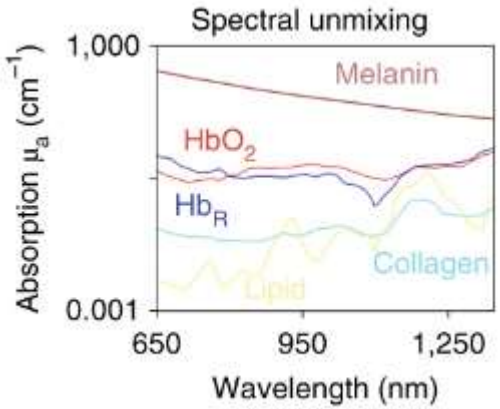
Optoacoustics could quantify haemorrhages (hemoglobin) and lipids in a atherosclerotic plaques in the carotid artery

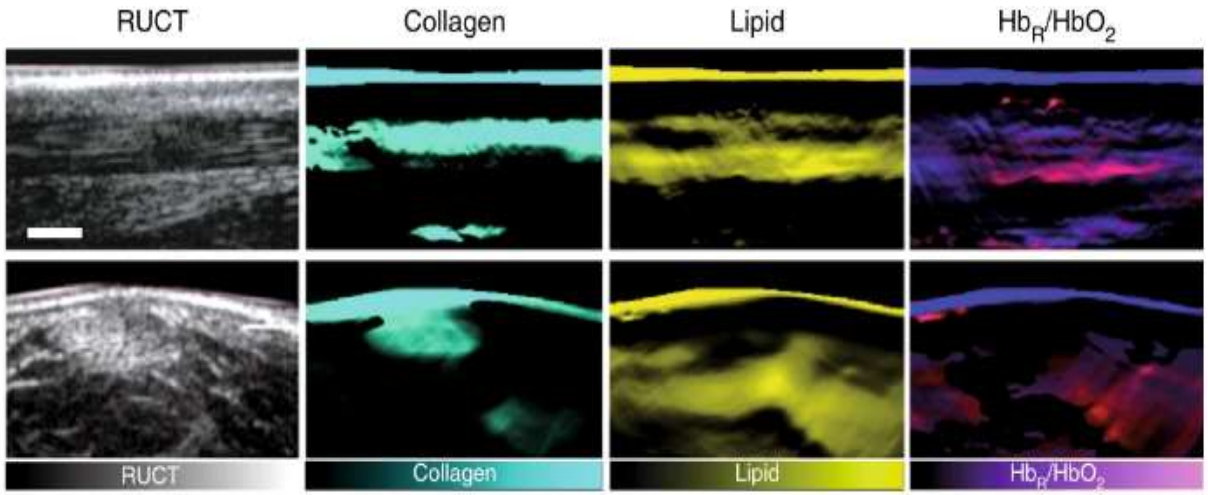


Muscular dystrophy

Duchenne muscular dystrophy (DMD) is a genetic disorder characterized by progressive muscle degeneration and weakness

Optoacoustics enabled quantifying the collagen content, which correlated with the functional status of patients





Take-home messages

- Ultrasound remains a cornerstone of clinical imaging, offering distinct advantages in medical diagnostics

- A new era of ultrasound is unfolding with the development of new methods that are transforming imaging performance

- Optoaocustic imaging is growingly used in preclinical research, including studies in oncology, neuroscience, cardiovascular biology, and other fields

- Recent initial clinical trials have demonstrated that optoacoustic imaging has potential for clinical translation

And that was it for today

Thank you